A continental-scale analysis of genomic variation in a songbird

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Abstract

A major aim of evolutionary biology is to understand why patterns of genetic variation vary within genomes, and among populations and species. Widespread continental species are interesting models for studying how evolution shapes patterns of genomic divergence, as they are associated with recent and rapid colonisations, large population sizes and high levels of gene flow. We studied patterns of genomic variation among 28 great tit (*Parus major*) populations in Europe. We found that genome-wide variation was consistent with a recent colonisation across Europe from a single refugium in the Balkans and/or Turkey. The highest levels of genome-wide structure were found between island populations in Crete, Sardinia and Corsica, most likely as a result of mild bottlenecks and reduced gene flow among these populations. We then found that the landscape of genomic differentiation was highly heterogeneous, with clear "islands of divergence", even among ostensibly panmictic great tit populations. Islands of divergence were consistently found in regions of low recombination, even among the most closely related populations. suggesting that background selection can rapidly promote population differentiation among large, recently colonised populations undergoing gene flow. Finally, we found that the relationship between recombination and the landscape of genomic differentiation mirrored the population history of great tits in Europe, with the strongest relationship between population divergence and recombination found among the most distantly related populations. Together, our findings suggest that studies of genomic divergence in the wild should consider population history, demography and genomic architecture, as well as likely patterns of natural selection.

Introduction

Since the first studies of allozyme variation in humans (Harris, 1966) and *Drosophila* (Hubby & Lewontin, 1966; Lewontin & Hubby, 1966), there has been great interest in explaining how evolutionary and ecological processes shape the patterns of genetic variation observed within and among natural populations. One focus of research and debate in this area has been on quantifying the roles of adaptive and neutral processes in explaining observed levels of genetic variation (Nei, 2005). However, adaptation does not occur in isolation, but against a background of recombination, gene flow, and genetic drift. More recently there has been increased effort in understanding how these fundamental evolutionary forces operate in concert to generate and maintain the levels of genetic diversity commonly observed in natural populations (e.g. Feder, Egan, & Nosil, 2012; Lanfear, Kokko, & Eyre-Waler, 2014).

The arrival of high-throughput sequencing and subsequent characterisation of genome-wide variation across large numbers of individuals has revealed that at the genetic level, patterns of variation and divergence among

natural populations and species are highly heterogeneous (Turner, Hahn, Nuzhdin, Mitchison, & Spencer, 2005). A feature of these "genomic landscapes" of differentiation that has received particular attention is the presence of so-called "islands of divergence"; outlier regions of the genome with high levels of divergence estimated from statistics such as F_{ST} or d_{xy} (N. J. Nadeau et al., 2011; Poelstra et al., 2014; Renaut et al., 2013; e.g. Turner et al., 2005). Initially these regions were termed "islands of speciation", and were thought to arise as a result of reduced gene flow in genomic regions associated with reproductive isolation (Turner et al., 2005). Subsequent modelling and analysis revealed that highly heterogeneous patterns of genomic divergence can occur in the complete absence of gene flow, as a result of linked selection on beneficial or deleterious mutations (Cruickshank & Hahn, 2014; M. A. F. Noor & Bennett, 2009).

Comparing patterns of genomic differentiation among sets of populations and/or species at different stages of the speciation continuum is a powerful way of disentangling the forces that shape variation among populations. Martin et al. (2013) showed that, across multiple *Heliconius* butterfly populations and species, patterns of genomic variation were shaped by a combination of gene flow and selection, particularly in genomic regions harbouring genes involved in wing patterning. In contrast, Renaut et al. (2013) showed that in *Helianthus* sunflowers, genomic architecture was the main driver of genomic differentiation across sets of populations. Similarly, recent research in birds has revealed that differentiation landscapes are conserved across populations, species and even across avian families (Reto Burri et al., 2015; Van Doren et al., 2017; N. Vijay et al., 2017). This latter pattern appears to have arisen, at least in part, as a result of a highly conserved recombination landscape in birds.

It is now clear that the recombination landscape is a key driver of genomic variation, but we are only just beginning to understand how this recombination rate variation interacts with other evolutionary forces to shape patterns of differentiation across natural populations and species (R. Burri, 2017; H. Ellegren & Wolf, 2017; C. D. Jiggins & Martin, 2017; Lohse, 2017; Ravinet et al., 2017). A recent, large-scale analysis in sticklebacks showed that, like in many other studies, islands of differentiation were most likely to be found in regions of low recombination, but by looking at this relationship across many pairs of populations they showed that outlier genomic regions were more likely to arise in low recombination regions when gene flow occurred between populations (Samuk et al., 2017). There is also a significant impact of how recently separated populations are, and it is expected that in very recently separated populations the differentiation landscape is most likely to reflect selective sweeps, then as divergence accumulates, genomic architecture plays an increasingly important role (R. Burri, 2017). More studies of genomic differentiation across very recently separated populations are now required to examine i) how rapidly divergence can accumulate in low recombining regions, and ii) how selection, gene flow and genomic architecture interact to shape genomic

divergence in the early stages of the differentiation process.

The European great tit (*Parus major major*) is an excellent model for ecological and evolutionary study (Gosler, 1993). Phyleogeographic research using mitochondrial and microsatellite loci suggests that this species has recently expanded across its European range, possible from a single refugium in the Balkans (L Kvist, Ruokonen, Lumme, & Orell, 1999). Contemporary populations are characterised by large effective population sizes and high levels of gene flow among populations, resulting in low levels of genetic differentiation (Laura Kvist et al., 2003; Lemoine et al., 2016). The genome of the great tit has recently been sequenced (Laine et al., 2016), and a high density (650K) panel of SNP markers has been developed. A recent study of two European populations using this marker panel suggests that rapid adaptation has occurred at the genomic and phenotypic levels *(Bosse et al. 2017)*. A wealth of ecological data also exists across multiple great tit populations (Charmantier et al., 2008; Dingemanse et al., 2012; e.g. M. E. Visser, Noordwijk, Tinbergen, & Lessells, 1998), enabling us to generate informed hypotheses about selection in this system.

In this study we examined how selection, gene flow and population history have shaped patterns of genomic variation across recently colonised European great tit populations. Using a large SNP panel typed across 28 populations, we first characterise variation within and among populations, in order to infer patterns of population history. We then examine how variation is partitioned across the genome, and test the hypothesis that highly divergent genomic regions have arisen as a result of variation in recombination rate across the genome. Finally, we examine how the relationship between genomic divergence and recombination varies according to geography, with the aim of inferring how gene flow and drift interact with selection to shape genomic landscapes in natural populations.

Methods

Sampling and molecular methods

Samples were collected from 28 populations from 22 countries across Europe (Fig. 1; Table 1). Birds were trapped from nest boxes, or using mist nets, and ringed with a uniquely numbered aluminium ring. Blood was taken via brachial venipuncture, and stored in either 1 ml Cell Lysis Solution (Gentra Puregene Kit, Qiagen, USA) or Queen's buffer. All samples were genotyped using a custom-made Affymetrix® great tit 650K SNP chip at Edinburgh Genomics (Edinburgh, United Kingdom). After filtering, a total of 647 samples typed at 483888 SNPs were retained for analysis (Table 1).

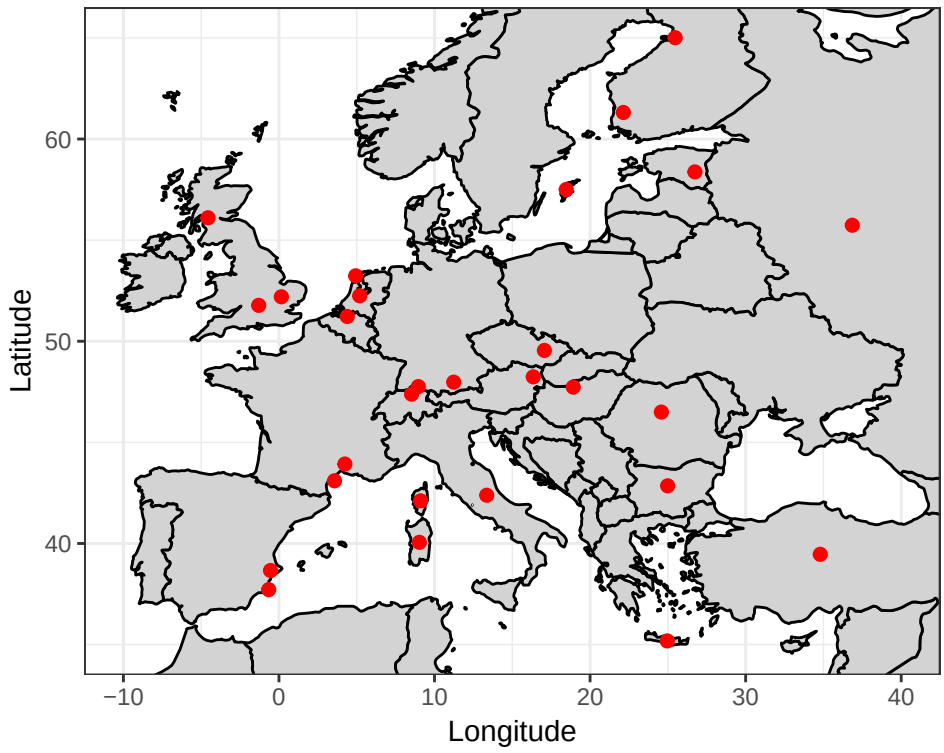

Figure 1 Sampling locations of great tit populations used in the present study. The size of the circles represents the extent of short term LD (markers within 1kb) within each population.

Analyses

Unless stated otherwise, all population genetic statistics were calculated in PLINK version 1.9 (Purcell et al., 2007), and downstream analysis and plotting was carried out in R version 3.3 (R Development Core Team, 2011). In each population, we estimated linkage disequilibrium (R^2) for each pair of markers within 50kb on the same chromosome, and compared this to physical distance between marker pairs. For a simple comparison of short-range LD among populations, we also calculated, for each population, mean LD between all pairs of markers within 1kb. We calculated observed heterozygosity for each SNP and population using a reduced SNP dataset, which was pruned based on linkage disequilibrium to remove all markers with $R^2 > 0.1$, then thinned with a probability of retaining each variant of 0.25.

We calculated genome-wide (mean) F_{ST} between each pair of populations using the pruned and thinned dataset described above. Pairwise F_{ST} was compared to geographic distance between populations using Mantel tests, implemented in the Ecodist package in R (Goslee & Urban, 2007). We tested whether genetic structure was related to distance from candidate refugial populations (in the Balkans, Turkey, Spain and Italy), using Pearson correlations. We also estimated population structure using Admixture version 1.3, with

default settings (Alexander, Novembre, & Lange, 2009). We varied values of K from one to ten; by which time increasing values of K provided no informative information about population structure (see results). Model support for each value of K was estimated by calculating 5-fold cross-validation error. Finally, we visualised the evolutionary history among European great populations by generating a maximum likelihood tree in TreeMix version 1.13 (Pickrell & Pritchard, 2012). We rooted the tree using a sample of *P. minor* individuals sampled from Amur, Russia. We fitted models allowing for range of migration events (0-10), and used a window size of 500 SNPs (Pickrell & Pritchard, 2012). To assess model fit, we calculated the proportion of variance in relatedness between populations explained by each model (equation 30 in Pickrell & Pritchard, 2012).

We examined the genomic landscape of differentiation across European great tit populations by calculating F_{ST} in 500kb bins, using python scripts obtained from Github (https://github.com/simonhmartin/genomics_general). We did not use estimates d_{xy}, as this parameter is difficult to estimate accurately from single SNP loci (Cruickshank & Hahn, 2014). We also calculated standardised F_{ST} (hereafter ZF_{ST}) by mean-centring windowed values and dividing them by the standard deviation among windows. We defined outlier regions as 500kb bins with zF_{ST} values greater than ten. We tested whether the landscape of genomic differentiation was related to recombination rate variation using a recombination map previously developed for the great tit using a 10K SNP chip (Oers et al., 2014), from which we estimated recombination rates in 500kb non-overlapping windows.

To gain further insight into how selection may have shaped variation in outlier regions, we examined patterns of haplotype variation. MIRTE TO ADD.

Results

Genetic diversity and population history

Sampling locations and sample sizes for each population are given in Table 1. Levels of genetic diversity were generally high and LD declined rapidly in all populations, but we observed substantial differences in genetic diversity among populations. Highest levels of LD were observed in the island populations of Crete (Greece), Sardinia (Italy), Corsica (France) and Vlieland (Netherlands) with lowest levels of LD in central Europe, Turkey and Spain (Fig. S1). Similarly, heterozygosity was highest in central Europe, and lowest in Crete, Sardinia and the Balkans (Fig. 2A).

Considering all European great tit populations, there was no significant pattern of isolation-by-distance (Mantel test; r = 0.14, p = 0.16). Instead, the highest levels of F_{ST} were found in comparisons involving the

island populations of Corsica, Sardinia and Crete (Fig. 2A Fig. 2B). Considering only comparisons involving mainland populations, pairwise F_{ST} was still not related to related to geographic distance (r = 0.07, p = 0.33). We then examined genetic structure in relation to distance from potential refugial populations (only including mainland populations), and found that F_{ST} was significantly related to distance from Turkey (r = 0.41, p 0.043; Fig. 2C) and the Balkans (r = 0.19, p 0.193), but not to distance from Spain (r = -0.26, p = 0.07), or from Italy (r = -0.17, p = 0.41).

Admixture analysis revealed patterns of genetic structure corresponding to geographical variation in European great tits (Fig. S2). The $K = 2$ analysis assigned individuals in Sardinia and Corsica to one genetic cluster, and the remaining populations to the second. At $K = 3$ (the model that best fitted the great tit data; Fig. S3), Spain was separated from the rest of mainland Europe. Increasing values of K resulted of the separation of populations in Scotland ($K = 4$), Sardinia (from Corsica; $K = 5$), France ($K = 6$), Crete ($K = 7$) and England ($K = 8$). The Admixture output at $K = 8$ is displayed in Fig. 2C; further increases in K did not generate patterns of structure that corresponded to geographical variation (Fig. S2), and were increasingly less well supported (Fig. S3).

Maximum likelihood analyses implemented in Treemix showed that a model with no migration explained 97.8% of variance in relatedness between populations; increasing the number of migration events substantially improved the percentage of relatedness explained, up to 99.7% when 10 migration events were fitted (Fig. S4). In Figure 3 we display the maximum likelihood tree with 2 migration events, after which the variance in relatedness explained plateaued when more migration events were added (Fig. S4). The tree was generally characterised by short branch lengths, with the exception of the island populations of Sardinia and Crete, which were grouped with the population from mainland Italy. The Spanish and Corsican populations were also grouped together, as were populations from Turkey and the Balkans, and populations from England and Scotland (Fig. 3). The two fitted migration edges both involved Sardinia, involving migration from eastern Europe, and to Corsica (Fig. 3).

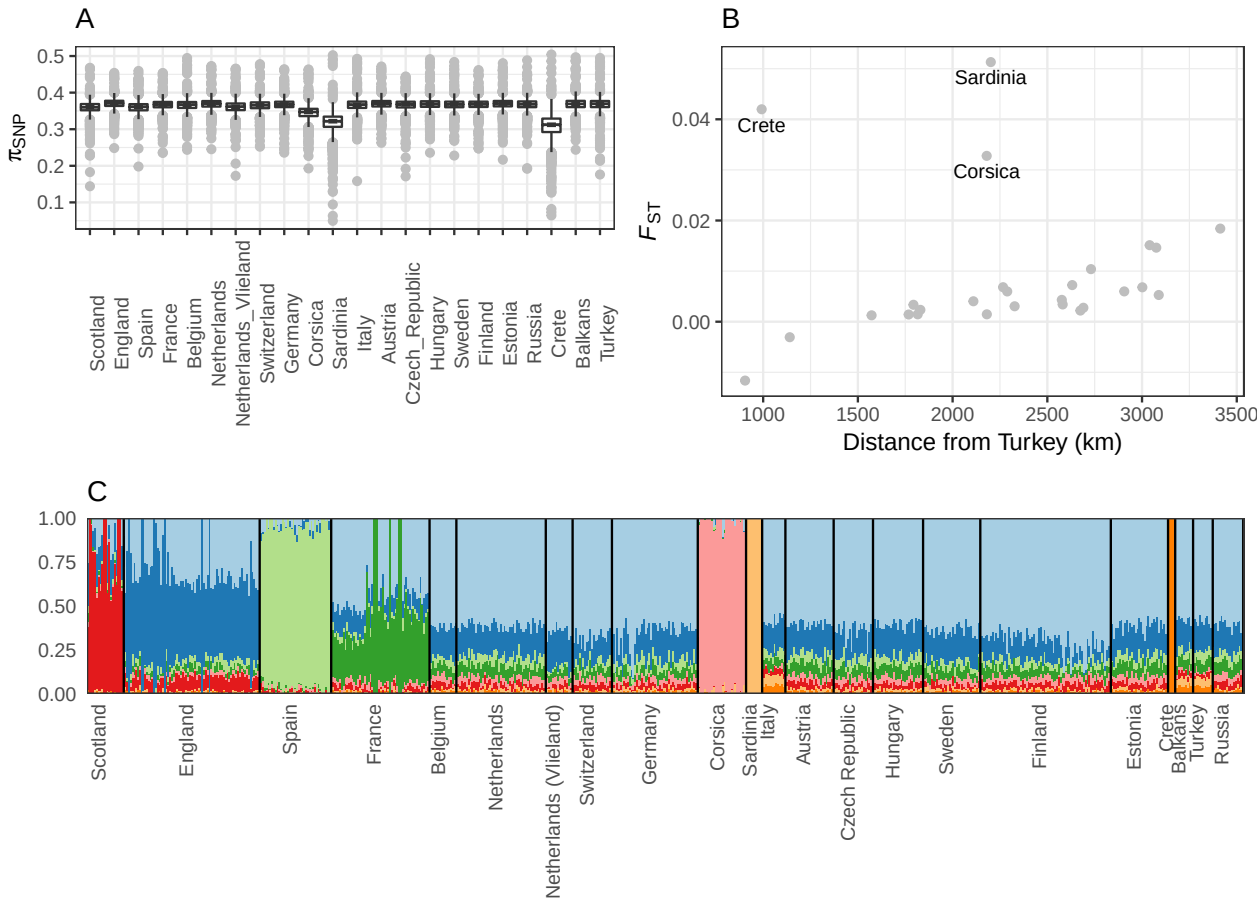

Figure 2 Genetic diversity and structure in European Great tit populations. **A** Nucleotide diversity within each population. **B** Pairwise F_{ST} in relation to geographic distance from the Turkey, only including comparisons involving Turkey. **C** Output from admixture analysis at $K = 8$.

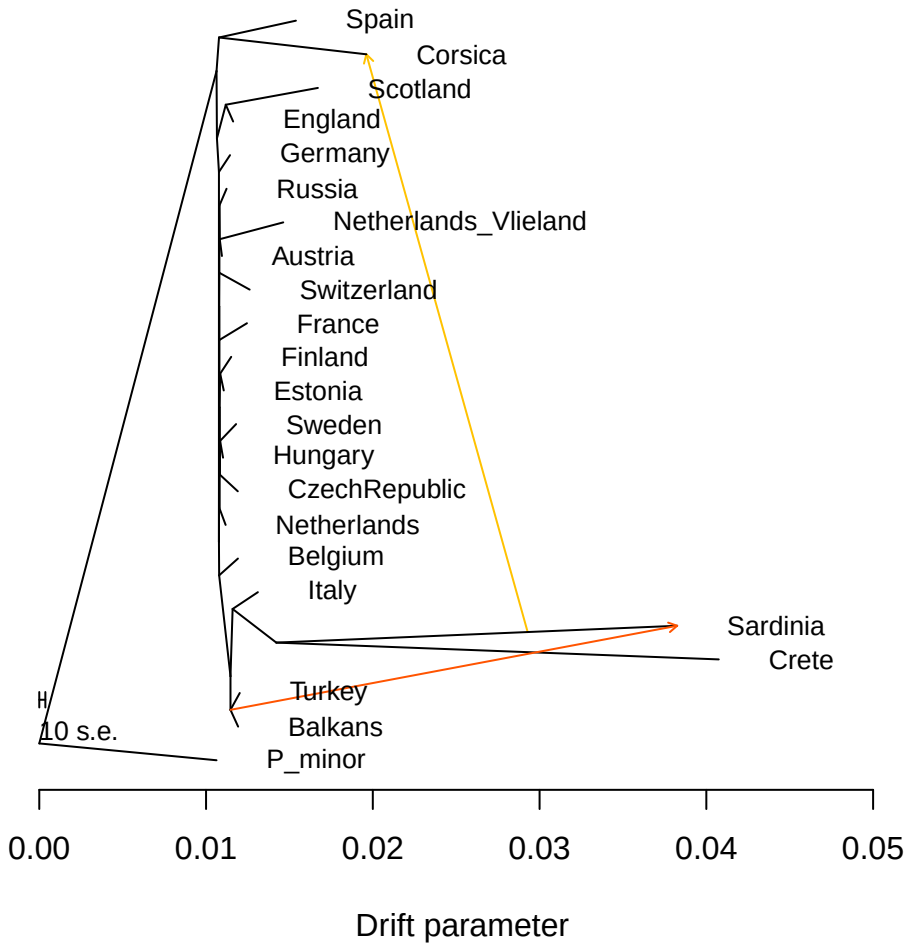

Figure 3 Maximum likelihood tree inferred by Treemix, allowing ten migration events. Migrations events (arrows) are coloured according to their weight (red = higher migration), and horizontal branch lengths are proportional to the amount of genetic drift that has occurred along the branch.

Genomic landscapes of differentiation

We examined the landscape of genomic divergence between each population and Turkey, and found that F_{ST} varied markedly across the genome in all comparisons (Fig. 4; Fig. S5). Outlier regions (windows with $zF_{ST} > 10$) were found in all comparisons apart from Crete and Sardinia, where overall levels of divergence were highest (Fig. 4). Of the 11 outlier regions, nine were found in only one or two comparisons, while the other two were found in 12 and 10 comparisons, respectively (Table S2).

Genomic differentiation with Turkey was negatively related to recombination rate in almost all populations (Fig. 5). The relationship between F_{ST} and recombination rare was generally weak, with correlation coefficients < 0.1. In a handful of populations this relationship was substantially stronger - most notably in the island populations of Corsica, Sardinia and Crete, and in England, Scotland and Spain (Fig. 5). Outlier loci of very high high differentiaion almost exclusively occurred in relatively low recombining regions, and we found no evidence that "shared" outlier regions were more likely to be found in low recombining regions compared to loci that were outliers in only one or two populations. The recombination rate of outlier regions was significantly lower than the genome-wide average (Wilcoxon test, $P < 0.001$), and there was no difference in recombination rate between 'shared' and 'unique' outlier regions ($P = 0.29$).

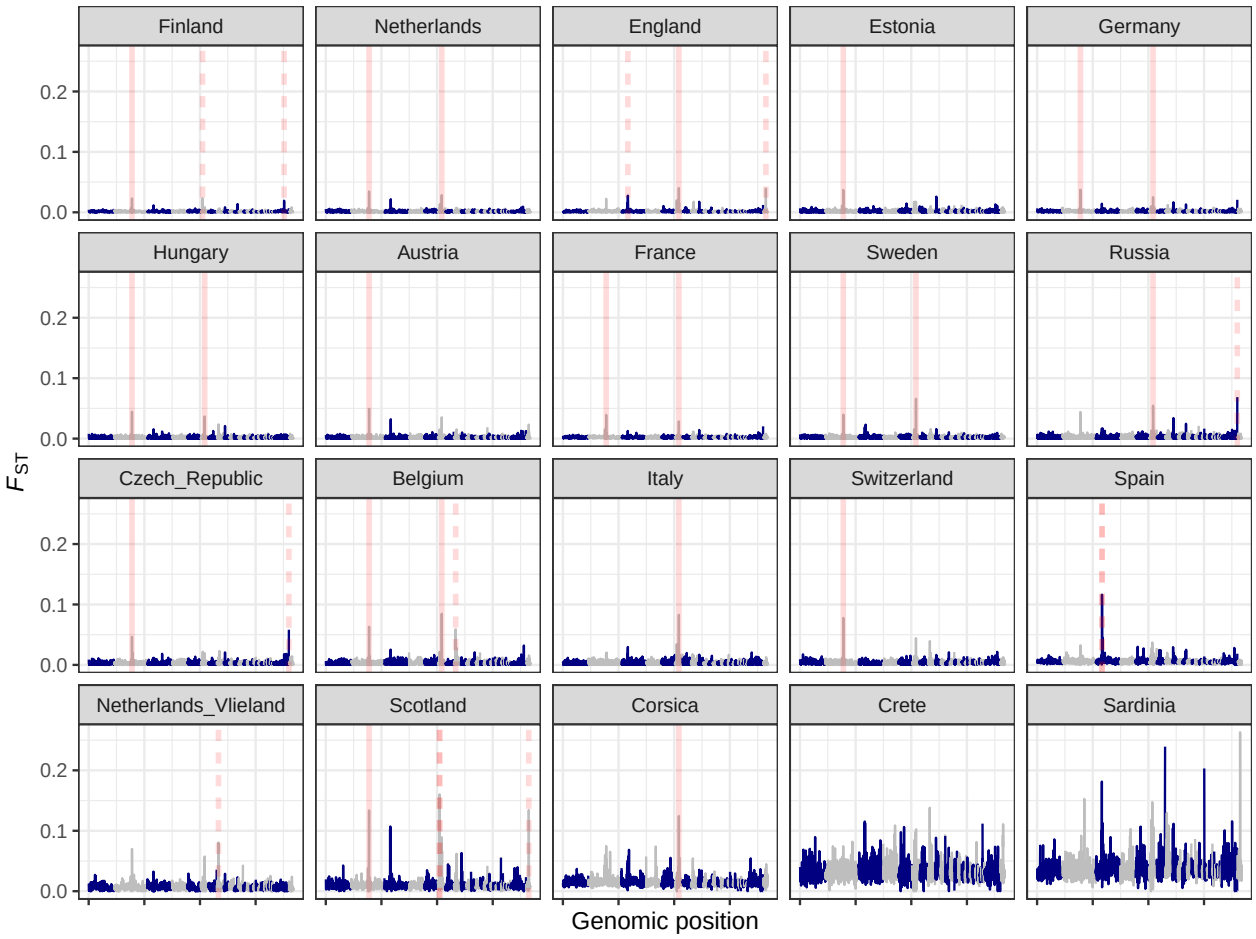

Figure 4 Landscapes of genomic differentiation in European great tit populations. F_{ST} across the genome is averaged in 500kb windows, with each panel displaying a pairwise comparison between with the proposed refugial population in Turkey. Red lines represent F_{ST} outliers (windows with mean F_{ST} values at least 10 standard deviations greater than the global mean for that comparison) shared across more than two comparisons (solid red lines), or unique to one or two comparisons (dashed red lines).

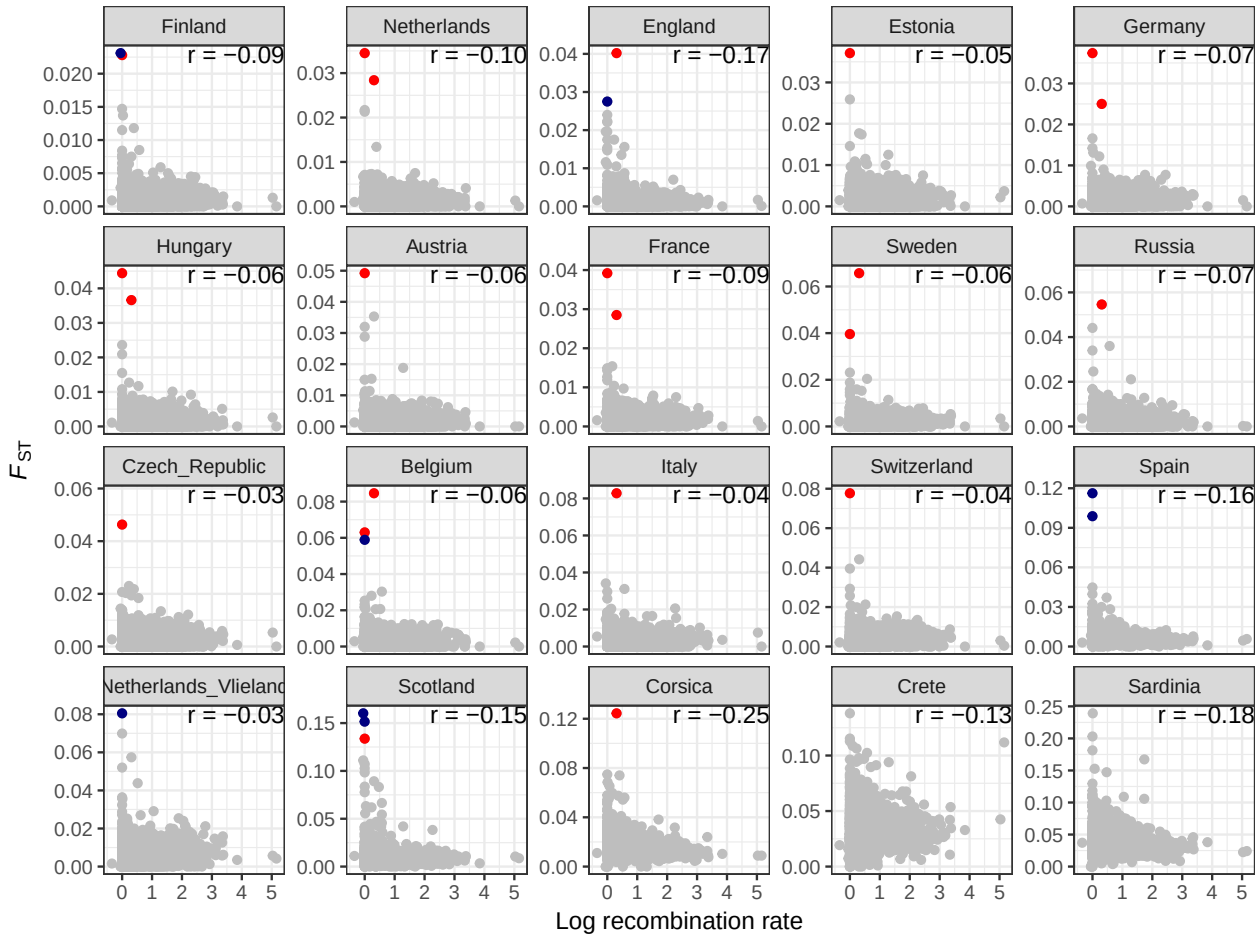

Figure 5 Genomic differentiation and recombination rate variation in European great tit populations. Each point is a 500kb window, with each panel displaying a pairwise comparison between with the proposed refugial population in Turkey. Coloured points represent F_{ST} outliers (mean zF_{ST} values > 10) shared across more than two comparisons (red), or unique to one or two comparisons (navy).

Discussion

In this study we use a large panel of genetic markers to provide insight into the evolutionary history of the great tit in Europe. We found that, consistent with previous research in this species, levels of genomic variation were consistent with a large effective population and substantial levels of gene flow among populations (Laura Kvist et al., 2003; Laine et al., 2016). We then examine how genomic landscapes of differentiation vary among different sets of populations, and find that i) islands of differentiation arise among extremely closely related populations, ii) these same islands of differentiation arise in more distantly related populations, iii) islands of differentiation arise in regions of low recombination, iv) the relationship between recombination rate and the genomic landscape of differentiation varies substantially among populations. These findings have several implications for how we interpret patterns of genomic variation among natural populations.

Our results suggest that great tits colonised Europe from a single refugium in Turkey and/or the Balkans. This scenario has been suggested before (L Kvist et al., 1999), and our observed pattern of increasing genome-wide divergence from the Turkish and Balkan populations lends this hypothesis empirical support. Genomic structure among mainland populations is, however, extremely low, and even with hundreds of thousands of markers Admixture was unable to separate the central European populations. Similarly, in the Treemix analysis the central European great tit populations formed a large group characterised by shallow branching, although much of the grouping that did occur (e.g. Finland and Estonia) made geographical sense. Interestingly, Treemix grouped the Spanish and Corsican populations, which is consistent with previous subspecies descriptions of European great tits (REF). More generally, the highest levels of differentiation were found between island populations, suggesting a reduced effective population size and reduced gene flow in these populations.

It is likely that the many, and perhaps the majority, of wild populations are characterised by highly heterogeneous patterns of differentiation across the genome (see REFS for a recent overview). What is not yet clear is how quickly "islands" of differentiation arise, nor how selection, gene flow and drift interact with genomic architecture to shape differentiation over shorter and longer evolutionary time-scales (R. Burri, 2017). Our results suggest that islands of differentiation arise in regions of low recombination even among populations that could be considered panmictic. The central European great tit populations have a mean F_{ST} of 0.005, yet islands of divergence can clearly be seen, and occur almost exclusively in regions of low recombination, regardless of which populations were examined (Figs 4,5). Thus, our results suggest that linked selection in regions of low recombination plays a key role in driving patterns of genomic differentiation even in the very earliest stages of population separation. This is not necessarily what we would expect - in the

very earliest stages of differentiation it is often assumed that peaks of high differentiation are the product of selective sweeps, and that only after time do correlated patterns of genomic divergence arise (R. Burri, 2017). It is difficult to differentiate between positive and background selection in producing heterogeneous patterns of differentiation across the genome. Correlated patterns of genomic differentiation among sets of populations suggests that background selection likely plays a role (REFS), and that is what we find in great tits.

Patterns of increasing genetic differentiation with distance from In accordance with this, we find that the tendency for outlier regions to occur in regions of low recombination increases with geographical and genetic distance from the refugial population in Turkey (Fig. 5). Thus we suggest that in continental species, the increased genome-wide divergence that accumulates with distance from refugial populations is accelerated in regions of low recombination, resulting in highly heterogeneous landscapes of genomic differentiation.

Needs more. Perhaps a discussion on role of gene flow... also need to think about what additional analyses may be required to strenghtn/add novelty to the story. One thing we could do is look in more detail at how genomic differentiaiton builds up with distance from the refugial population? Thoughts?

Acknowledgements

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